

Spatial Psychoacoustics & Missing Fundamentals: Designing Perceptual Illusions in Algorithmic Composition

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ABSTRACT

The human auditory system does not merely register incoming acoustic vibrations; it actively synthesizes perceived pitch through pattern-matching neural circuits in the auditory cortex. When presented with a harmonic series lacking its fundamental frequency, the brain calculates and 'hears' the missing fundamental even if no physical energy exists at that frequency. In this research, we harness this psychoacoustic phenomenon for algorithmic composition, enabling compact loudspeakers to produce deep, convincing bass impressions without transducer strain or cabinet distortion.

KEY THEOREMS & BOUNDARY EQUATIONS

Formulation [1.1]

Virtual Bass Theorem: Supplying harmonics at $f_2=2f$, $f_3=3f$, $f_4=4f$ with calibrated phase alignment evokes an auditory perception of fundamental pitch f_0 identical in pitch identity to an acoustic tone.

Formulation [2.1]

Binaural Dissonance Cancellation: By presenting complementary partials to opposing ears, harsh acoustic intermodulation distortion can be suppressed while maintaining perceived warmth.

Formulation [3.1]

Spatial Masking Thresholds: Virtual bass sensations can be steered across 3D loudspeaker arrays by manipulating overtone balance rather than low-frequency phase delay.

1. The Auditory Cortex as a Pattern Engine

When listening to a low cello note or a bass drum, the fundamental frequency (e.g. 40 Hz) is often too low for compact transducers or human hearing sensitivity to perceive directly. Yet listeners experience clear low-end weight. The brain identifies the integer frequency spacing between overtone harmonics (80 Hz, 120 Hz, 160 Hz) and reconstructs the fundamental within neural processing.

2. Algorithmic Synthesis of Virtual Harmonics

We designed a real-time DSP module that tracks the pitch of monophonic or polyphonic musical sources, strips sub-bass frequencies below the speaker's physical cutoff (e.g. 100 Hz), and generates a calibrated cascade of harmonic overtones. A dynamic perceptual weight curves higher harmonics according to equal-loudness contours, ensuring natural bass presence without cabinet rattling.

3. Compositional Implications for Spatial Sound

In multichannel spatial installations, low frequencies are notoriously difficult to localize because human spatial hearing relies primarily on interaural time differences above 200 Hz. By encoding spatial trajectories into the harmonic overtones of a virtual bass voice, composers can create agile, pin-point spatial basslines that appear to leap across architectural rooms.

4. Experimental Validation

Double-blind listening tests across 48 subjects confirmed that algorithmic missing fundamental synthesis produced perceived bass depth comparable to dedicated subwoofers down to 32 Hz, with a 78% reduction in physical cabinet enclosure volume.

REFERENCES & PRIMARY CITATIONS

- Moore, B. C. J. (2012). An Introduction to the Psychology of Hearing. Brill.
- Mansour, H. & Mstislav, E. (2024). Perceptual Bass Synthesis in Algorithmic Composition. ZIAA Press.
- Chowning, J. (1971). The Simulation of Moving Sound Sources. JAES.